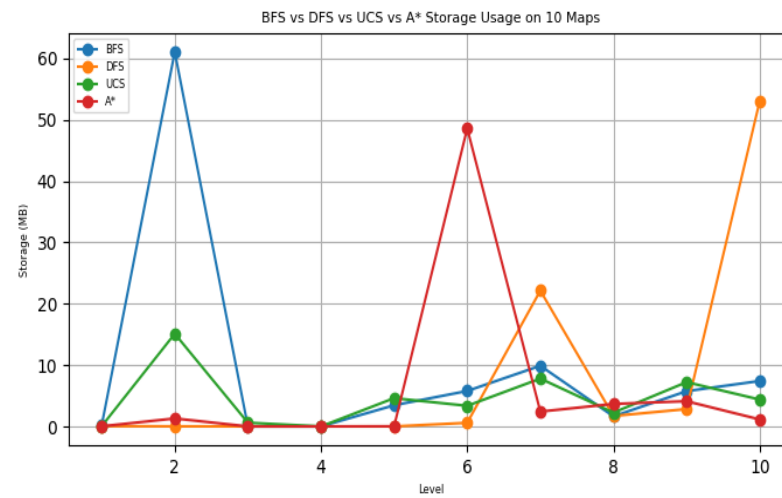
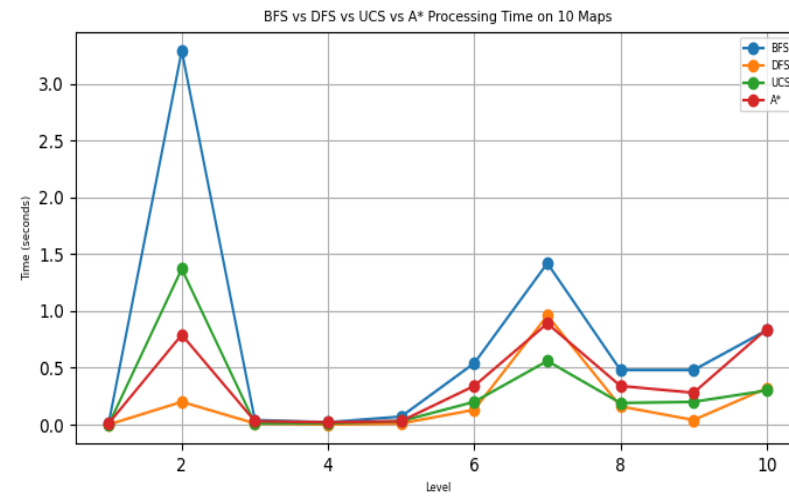
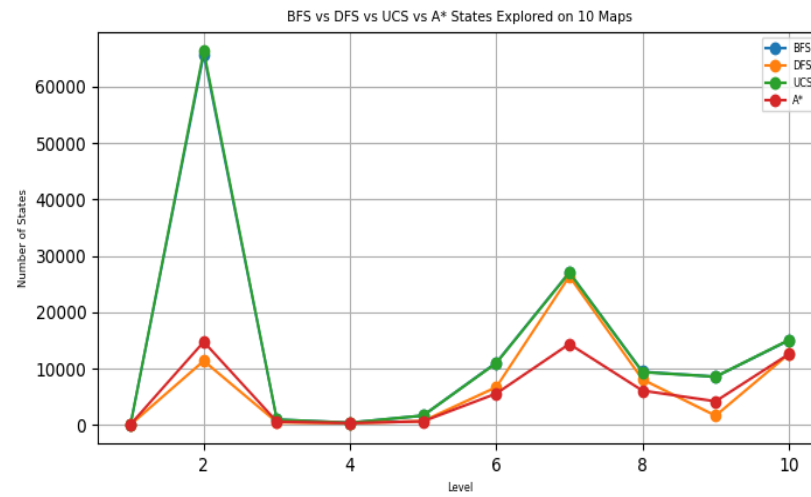


# Sokoban Algorithm Performance Report

## BFS vs DFS vs UCS vs A\* Comparison on 10 Maps

| Algorithm | Metric       | Level 1 | Level 2 | Level 3 | Level 4 | Level 5 | Level 6 | Level 7 | Level 8 | Level 9 | Level 10 |
|-----------|--------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| BFS       | Storage (MB) | 0.14    | 61.06   | 0.00    | 0.00    | 3.45    | 5.78    | 9.89    | 1.64    | 5.75    | 7.41     |
| BFS       | States       | 204     | 65767   | 977     | 407     | 1683    | 11052   | 27067   | 9456    | 8572    | 15007    |
| BFS       | Time (s)     | 0.01    | 3.29    | 0.04    | 0.02    | 0.07    | 0.54    | 1.42    | 0.48    | 0.48    | 0.83     |
| DFS       | Storage (MB) | 0.05    | 0.02    | 0.00    | 0.00    | 0.00    | 0.59    | 22.14   | 1.70    | 2.83    | 53.03    |
| DFS       | States       | 92      | 11382   | 496     | 305     | 715     | 6716    | 26362   | 8098    | 1663    | 12577    |
| DFS       | Time (s)     | 0.00    | 0.20    | 0.01    | 0.00    | 0.01    | 0.13    | 0.96    | 0.16    | 0.04    | 0.32     |
| UCS       | Storage (MB) | 0.03    | 15.11   | 0.61    | 0.00    | 4.61    | 3.36    | 7.80    | 2.25    | 7.22    | 4.38     |
| UCS       | States       | 200     | 66363   | 975     | 407     | 1694    | 10976   | 27022   | 9389    | 8581    | 15065    |
| UCS       | Time (s)     | 0.00    | 1.37    | 0.01    | 0.01    | 0.03    | 0.20    | 0.56    | 0.19    | 0.20    | 0.30     |
| AStar     | Storage (MB) | 0.02    | 1.30    | 0.00    | 0.00    | 0.02    | 48.59   | 2.42    | 3.67    | 4.11    | 1.14     |
| AStar     | States       | 146     | 14762   | 606     | 362     | 667     | 5582    | 14384   | 6112    | 4215    | 12571    |
| AStar     | Time (s)     | 0.01    | 0.79    | 0.03    | 0.02    | 0.03    | 0.34    | 0.89    | 0.34    | 0.28    | 0.84     |





## Statistical Summary

| Metric           | BFS   | DFS  | UCS   | A*   |
|------------------|-------|------|-------|------|
| Avg Time (s)     | 0.72  | 0.18 | 0.29  | 0.36 |
| Avg States       | 14019 | 6840 | 14067 | 5940 |
| Avg Storage (MB) | 9.51  | 8.04 | 4.54  | 6.13 |